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Intelligent air traffic control using NLP-enhanced speech recognition and natural language generation

Amany M. Sarhan[1*], Rawda Fathy[2] and Hesham A. Ali[2,3]

*Correspondence: amany_sarhan@f-eng.tanta.edu.eg

[1] Department of Computer and Control Engineering, Faculty of Engineering, Tanta University, Tanta, Egypt
[2] Department of Artificial Intelligence, Faculty of Artificial Intelligence, Delta University, Gamasa, Egypt
[3] Department of Computer and Systems Engineering, Mansoura University, Mansoura, Egypt

Abstract

The combination of artificial intelligence (AI) and natural language processing (NLP) can bring an intelligent solution to air traffic management (ATM) for reliability, accuracy, and safety. This research aims to present a real-time intelligent system that improves the communication between air traffic controllers (ATCOs) and pilots. The proposed system enhances transcription accuracy, supports automated decision-making, reduces the response time and furthermore improves safety in high-air traffic situations. The proposed architecture is designed to enable scalable integration of AI tools and NLP technologies in ATC systems. The merits of the proposed system is that it automates the whole steps of the communication in the air traffic system, rather than implementing parts of the process. The system begins by automatic speech recognition (ASR) module that is responsible for transforming the speech (instructions) into text. The architecture enables understanding the main ATC instructions via a specific list of most common keywords through a natural language understanding module (NLU), thus enabling the pilot to communicate with the ATM system. Additionally, the system incorporates natural language response generation (NLG) module to reduce pilot workload and improve the communication efficiency. Extensive experiments were conducted for system verification on several datasets, where each dataset is targeted for a specific module of the system. Through the results, the proposed system demonstrates its capability of reducing the communication errors and improving service reliability with the overall word recognition accuracy by the ASR module of 91.73%, while the NLU module gives F1 score to 0.9816, and the NLG module produced acceptable quality of generated texts (85%) and with generation latency ~ 0.6 s. Compared to the recent existing systems, our system gives a better overall performance.

Keywords: Air traffic control, Speech communication, Natural language processing, Natural language generation, Automatic speech recognition, Spoken instruction understanding

Introduction

Reliability and safety are two important measures that must be prioritized in aviation in specific and transportation in general. Both AI and NLP can help enhance aviation by their integration transforming communication between pilots and air traffic controllers

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(ATCOs), marking a significant technological shift as traditional methods are replaced by advanced, data-driven approaches [43]. Historically, aviation relied on manual communication and navigation systems, but as air traffic increased, the need for automation and precision became evident [14, 20, 41–43]. AI and NLP technologies offer significant advantages by addressing miscommunication, ambiguous language, and the overwhelming volume of air traffic data [13, 16, 17, 23]. NLP facilitates real-time language translation, sentiment analysis, and structured data extraction, allowing ATCOs and pilots to communicate seamlessly and reduce errors [31]. Studies suggest that AI-powered systems can significantly reduce communication errors, improving both safety and operational efficiency [24, 34, 44].

The existing ATC communication systems suffer from high air traffic volume, human limitations, and the possibilities of misinterpretation. Real-time decision-making and managing multilingual communications pose challenges that AI and NLP can successfully handle. Global aviation authorities, including the FAA and ICAO, are actively exploring AI solutions to improve air traffic coordination, safety monitoring, and strategic planning [14, 20, 40].

The major limitation of the traditional ATC systems is they are time-consuming and reliance on human which increases the risk of delays and safety instances. Miscommunication is a major safety hazard, necessitating the critical need for real-time speech recognition and automated decision support. AI-based speech processing systems can more effectively evaluate, classify, and respond to air traffic communications, minimizing workload and freeing up human controllers to concentrate on crucial decision-making [31, 32]. Current research shows that AI-assisted speech recognition can reduce manual processing time from 12,700 to 405 secs, encouraging their usage in real-time situations [31].

To this end, this study proposes an AI and NLP-powered framework to enhance ATC communication accuracy, reduce errors, and improve air traffic operations. This study intends to redefine air traffic management by combining automatic speech recognition with cutting-edge NLP models, guaranteeing increased effectiveness, safety, and flexibility to meet evolving aviation needs. By combining AI and natural language processing, this study offers multiple unique insights on aviation technology and air traffic control. The main contributions of this paper are summarized as follows:

1. Introducing an Integrated Framework for ATC: The proposed framework integrates AI and NLP technologies, including human response and natural language processing systems. It contains two basic modules: (i) understanding ATC instructions which categorizes ATC instructions using keywords like as climb and descend assisting in identifying the pilot's instructions after transforming the speech into text through speech recognition model and promotes for an appropriate response, and (ii) natural language response generation to produce natural language responses to pilot communications, thus reducing pilot load and increasing communication efficiency.
2. Validating The Framework Modules: The results of each module are given along with the datasets used for training. The results are compared against SOTA where it proves the framework ability to achieve its various goals.

This paper is organized as follows. Sect. "Related work" describes the most relevant current studies in AI and NLP for aviation, properly. In Sect. 3, the study's methodology will be discussed. Experiments and consequences are depicted in section four. In addition, the remaining part of the section covers the outcomes and results for the aviation industry.

Related work

The aviation sector has remained committed to developing communication systems that reinstate safety as a crucial benchmark in its operations. In the past, they have shifted from fewer and simpler radio communications to digital technology, communication, and satellite communications when technology advanced. Despite technological advances, ATC systems still face challenges with real-time processing and a rapidly changing multicultural spectrum in global aviation communication is a well-known fact [6, 18].

Yi et al. [43] presented a case study of implementing ASR systems in air traffic to increase safety factor showing that such systems significantly reduce miscommunication and enhance reliability. They used both simulated real-time data to evaluate ASR system ability to understand the pilot's instructions. Zuluaga-Gomez et al. [44, 45] also presented a complete analysis of the ASR systems performance in air traffic communications by using a benchmark for ASR accuracy for air traffic control. The key features that may affect the ASR performance are identified including background noise, speaker accents, and technical expressions.

A study [30] has attained high performance of ASR systems of 95% accuracy. According to [32] and a NASA report, AI technologies have the potential to greatly help the ATC organization by increasing controller–pilot communication and reducing miscommunication, which in turn improves aviation safety. Current research primarily focuses on developing highly accurate and stable ASR systems. One proposed solution involves innovative signal processing techniques to address challenges such as noise interference and speech recognition in noisy environments like aircraft [27].

Yang & Huang [41] presented sophisticated NLP algorithms to recognize important features and decipher pilot–air traffic controller conversations. These algorithms aimed to increase communication clarity by removing noise. AI algorithms is used to enhance multilingual, real-time communication between controllers and pilots [2]. So as to reduce human error and improve communication during crucial stages like takeoff and landing, recent research advises employing NLP and deep learning models to perform translation and comprehension of ATC signals.

The work presented by Alla et al. [1] explored using chatbots in enhancing operational efficiency, providing passenger assistance, and supporting air traffic control. The authors examined the technological developments of chatbots to perform these tasks efficiently and analyze the benefits and challenges associated with their implementation in the air traffic sector. The work in [35] showed how a real-time collected corpora can improve the effectiveness and accuracy of ASR and TTS. Training the deep learning models of an actual real-time conditions and the language used in air traffic control increases their accuracy rather than a fabricated dataset. This attention to domain precise statistics is useful for building structures able to address the difficulties of ATC communications.

Kistan et al. [14] showed that integrating sentence-level language identification improves multilingual speech recognition in air traffic control, enhancing communication speed and accuracy by reducing errors. Lin et al. [20, 21] proposed an adaptive collaboration framework to strengthen human–AI interaction in ATC, enabling more effective teamwork between AI agents and human controllers. The incorporation of multilingual speech recognition in ATC systems demonstrates AI's ability to meet linguistic requirements, supporting diverse languages to reduce communication barriers in global aviation [25]. NLP applications in air traffic management planning (ATMP) assist in analyzing large volumes of communication data, uncovering patterns, and enhancing operational planning and prediction capabilities.

Overall, AI in ATC offers significant benefits in communication, prediction, and decision-making, while also posing challenges in safety, reliability, and regulatory compliance [28]. AI enhances air traffic safety and efficiency by predicting and managing conflicts through real-time data analysis, allowing controllers to prevent incidents proactively [10]. AI-driven systems also harmonize communication protocols across regions, reducing language and procedural discrepancies in international ATC, which improves collaboration and operational safety [24].

Improving speech signal quality is critical for ASR performance in noisy aviation environments. Techniques for noise suppression and signal enhancement increase ASR reliability, ensuring clear, accurate communication between pilots and controllers [36]. Advanced signal processing strategies further boost ASR accuracy by filtering background noise, maintaining effective communication in challenging conditions [44].

Semi-supervised learning can improve ASR systems by leveraging both labeled and unlabeled data, enhancing accuracy and robustness. A deep neural network model for Chinese ATC using semantic ontology demonstrated improved trajectory estimation and instruction generation in real-world scenarios [39]. Yang & Huang [41] showed AI's potential in improving trajectory prediction and ASR reliability through analysis of large ATC communication datasets.

Preprocessing strategies also impact NLP model performance, particularly for Arabic text categorization, helping optimize models for specific linguistic contexts [38]. Region-specific ASR enhancements, as explored in [7], enable systems to better adapt to local languages and cultures, improving communication effectiveness.

Customizing ASR for ATC use, as shown in [9], ensures systems meet aviation-specific needs like technical jargon handling, boosting communication clarity and operational safety. Lázaro et al. [19] applied various ML classifiers to incident reports, using performance metrics and hyperparameter tuning to identify patterns and contributing factors in aviation incidents.

New and Wallace [26] proposed an NLP-based approach for classifying aviation safety reports, focusing on reproducibility, required training data size, and safe deployment in business settings. Using 425 reports, they found that at least 60 samples per class are needed to consistently achieve high performance (F1 score > 0.90).

In summary, integrating AI and NLP into air traffic control offers significant benefits in communication, efficiency, and safety. However, successful adoption requires addressing aviation-specific challenges such as testing, regulatory compliance, and continuous innovation. Table 1 offers a thorough comparative analysis of different research on

Table 1 Comparative analysis of AI and NLP applications in aviation

Paper	Advantages	Disadvantages	Focus	Methodology	Key findings
Zuluaga-Gomez, et al. [44], (2023)	Large-scale dataset creation, improved speech recognition and understanding, reduced human error	High annotation costs, domain shifts between different ATC environments	Creating a large-scale dataset for ASR and NLP research in ATC communications	Dataset creation, annotation processes, application of ASR and NLP technologies	Improved efficiency and safety of ATC communications, large-scale dataset availability
Fan, et al. [9], (2024)	Improved recognition performance, handling multilingual communications, leveraging sentence-level language identification	Recognition failures in multilingual settings, the complexity of training models	Developing a two-stage multilingual ASR framework with sentence-level language identification	Training language identifier and recognition model, integrating SLIL module	7.50% relative improvement in character error rate, effective handling of multilingual speech
Yang & Huang [41], (2023)	Enhanced aviation safety, analysis of communication data, identification of potential safety issues	Ambiguity in language, limited training data, lack of multilingual support	A systematic review of NLP applications in aviation safety and future outlook	Literature review, analysis of NLP applications and their impact	The potential of NLP to enhance aviation safety, challenges in implementation
Fan & Pan [8], (2024)	Improved ASR performance with domain-specific adaptation, reduced complexity in training new models	The complex training process requires large amounts of data and algorithms	Adapting ASR models to the ATC domain using improved fusion methods	Transfer learning, training RNN-based ASR model, improved shallow fusion	Improved ASR performance in ATC with reduced complexity, better language model performance
Šmídl et al. [35], (2019)	Dedicated speech corpora creation, enhanced training simulations	High costs and effort in corpus creation, specific pronunciation challenges	Creating speech corpora for ASR and TTS in ATC	Corpus creation, annotation, ASR, and TTS model training	Effective for training simulations, realistic pseudo-pilot interactions

the use of AI and NLP in the aviation field. Every study is assessed according to its plus points, minus points, target areas, the techniques used, and the important discoveries. The studies cover all areas of safety and efficiency in airspace management of communication systems and predictive maintenance; thus, they reflect the vast range of AI and NLP applications in aviation operations**.**

In conclusion, there are several benefits of integrating AI and NLP into air traffic control (ATC) systems, including increased situational awareness, quicker communication, and better decision-making. While AI offers real-time decision assistance by analyzing massive datasets to identify conflicts and optimize traffic flow, natural language processing (NLP) can automate the interpretation of pilot–controller conversations, freeing up controllers to concentrate on strategic duties. These technologies can improve airspace efficiency, decrease delays, and streamline operations. Their implementation also brings with it important problems, such as safety, dependability, and ethical issues, which need to be properly controlled to enable effective and secure adoption. A significant obstacle in integrating AI into air traffic control (ATC) is guaranteeing the algorithms' precision, dependability, and preparedness for the actual world. Confirming their efficacy in dynamic aircraft situations requires extensive testing. Furthermore, because ATC communications are confidential, rigorous privacy and cybersecurity protections, safety requirements, and regulatory compliance are essential.

Proposed framework

A strategic implementation plan that prioritizes real-time data processing, regulatory compliance, and smooth interface with current ATC systems is put out in order to address obstacles and fully utilize AI and NLP in aviation. Collaboration with authorities, ongoing algorithm development, redundancy procedures, and thorough testing to guarantee dependability and safety are essential elements.

Additionally, the plan asks for operational and cultural changes across the industry, backed by change management and training. Furthermore, automating pilot answers with natural language generation (NLG) improves communication effectiveness and security. For continuous improvement and long-term success, real-world feedback and performance evaluation will be essential.

The proposed framework is shown in Fig. 1 where ATC communication count on automated voice interactions. This study investigates how AI might be used to improve safety and communication clarity when flying. We suggest an architecture that uses NLP to extract commands, ASR to translate pilot vocal commands to text, and NLG to create understandable ATC responses. This AI-powered strategy seeks to increase overall ATC efficiency, reduce misunderstandings, and improve clarity. Currently, researchers are developing ways to automatically control pilot–air traffic controller communication. But each study just looks at one aspect of the air control cycle. The interoperability of different subsystems may be jeopardized, and the overall system designer may be burdened with choosing the right tool for each component and figuring out how to integrate these subsystems to produce the overall process image.

The new integrated air traffic control (ATC) framework presented in this study is depicted in Fig. 2, which shows the basic components of the system and how they interact with each other. The proposed ATC system consists of five modules, which are:

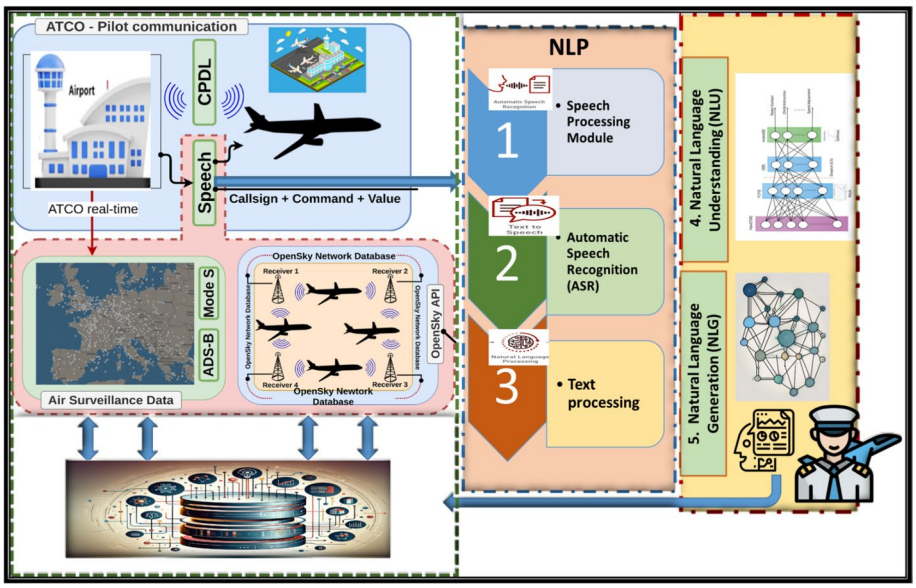


Fig. 1 Proposed AI-based aviation communication and surveillance system

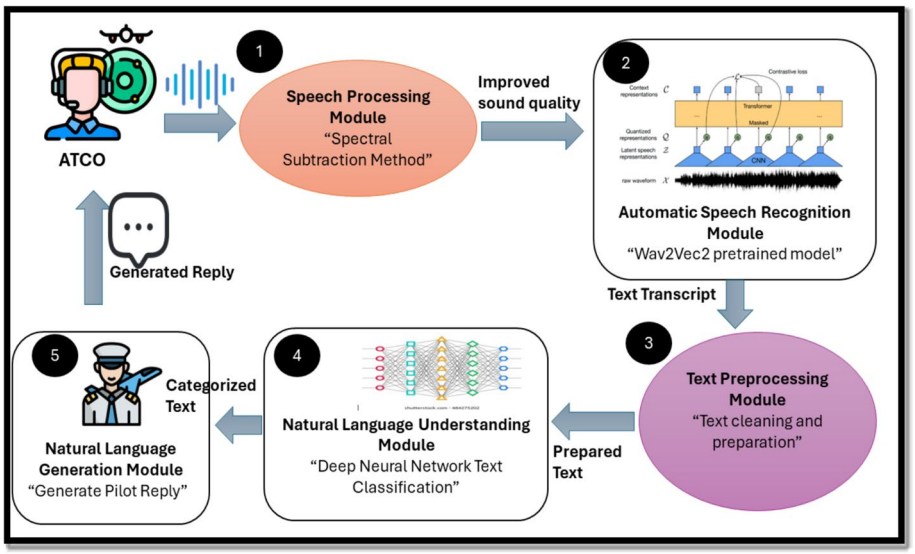


Fig. 2 Integrating ASR and NLP system for automated communication processing

(1) Speech Processing Module: Its role is to implement the enhancement of the ATC communication speech quality using a spectral subtraction method. a noise reduction approach that increases audio clarity by removing approximated background noise from the voice signal, to improve the quality of ATC communication.
(2) Automatic Speech Recognition Module (ASR): It converts ATC communication speech from ATCO to pilots to text using a deep learning pre-trained model.
(3) Text Preprocessing Module: It is responsible for preparing the raw text (from ASR module) for later analysis and processing in the NLU pipeline. The main goals of

text preprocessing are to decrease noise, normalize the text, and extract meaningful features usable by deep learning models.

(4) Natural Language Understanding Module (NLU): It categorizes instructions by their type to understand the intent behind the instruction, such as Radar Contact Instructions and Descend Instructions. This is done in two steps: Keyword-Based Labeling and Deep Neural Network Classification.

(5) Natural Language Generation Module (NLG): It is designed for generating text pilot replies, which are then refined by repositioning the airport names including the flight numbers, and removal of redundant ending phrases.

Speech processing module

Effective noise suppression ensures clear, background noise-free communications between pilots and air traffic controllers in critical areas like ATC. The accurate delivery of information and instructions, which is necessary for the safe working of airplanes, relies upon this clarity. When it comes to eliminating fixed noise—noise that persists over time, such background talk or engine hum—spectral subtraction works very well. This makes it appropriate for settings like aviation ATC where this kind of noise is common. The spectral subtraction approach is used in the proposed work to eliminate noise from voice communications between air traffic controllers and pilots.

As shown in Fig. 3, step 3, the spectral subtraction method aims to remove noise from a speech signal by estimating and subtracting the noise component from the signal's frequency-domain representation, and it is comprised of 7 main steps beginning after the speech signal input comes.

Fourier transform helps in analyzing a noisy signal in the frequency domain and accurately estimate and remove noise. The denoised signal is then reconstructed into the time domain for additional use using the inverse Fourier transform. Power spectrum

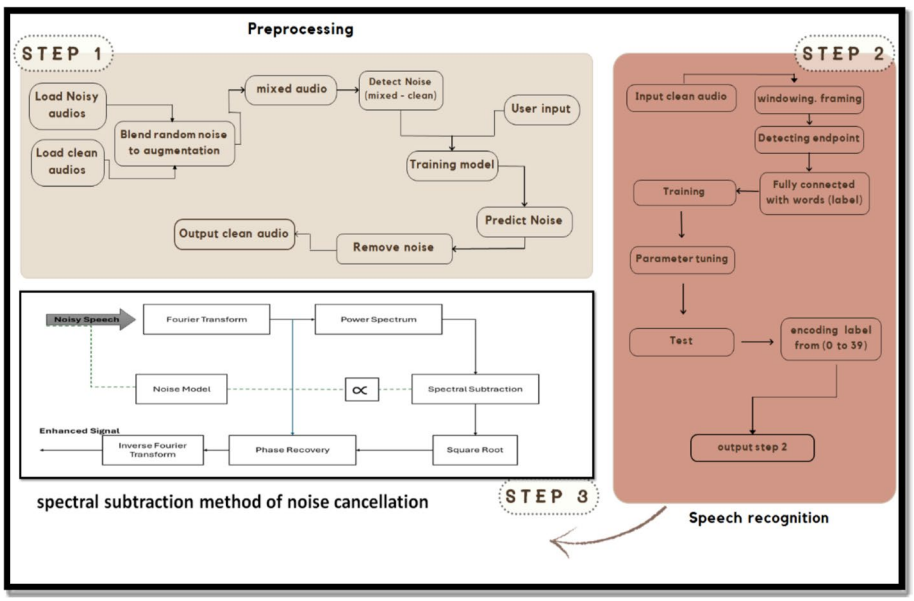


Fig. 3 Workflow of the spectral subtraction method of noise cancellation

helps us analyze the strength (energy) of the signal at different frequencies, by estimating the noise spectrum N(ω) and analyzing silent segments or using noise reduction techniques. Once we have the noise power spectrum, we can perform spectral subtraction (subtracting the estimated noise spectrum from the total signal power spectrum) to attenuate the noise component in the frequency domain. We can derive the power spectrum as follows:

N(ω) is an estimate of the noise spectrum that contaminates the voice signal. The precision of the noise model is crucial to the spectral subtraction method's efficacy. The noise model should ideally as precisely represent the noise's statistical characteristics as possible, including: (i) Noise is considered stationary if its statistical characteristics (mean, variance) remain constant throughout time. Because its spectrum is largely constant, spectral subtraction performs best when applied to stationary noise. (ii) Shape of the Spectrum: The spectrum shapes of various noise sources (such as white noise and power line hum) vary.

Based on the basic idea of spectral subtraction, α spectral subtraction reduces noise in audio signals by adding a scale factor (α) for optimization. Formula (1) is used to perform the subtraction.

$$Y(\tau, \omega) = X(\tau, \omega) - \alpha * N(\tau, \omega) \tag{1}$$

The scaling factor α is used to control the trade-off between noise reduction and signal distortion, where when α is closer to 1 will lead to a higher degree of noise reduction. Nevertheless, it increases the risk of subtracting signal components along with noise, leading to distortion when α is closer to 0.

By converting the modified spectrum obtained in the frequency domain back to the time domain. This allows us to recover the denoised signal for further processing or playback. The IFFT formula takes the modified spectrum Y(ω) and transforms it back to the time domain, resulting in the denoised signal y(t), as in (2).

$$y(t) = (1/2\pi) \int (-\infty, \infty)\ Y(\omega) * e^{\wedge}(j\omega t)\ d\omega \tag{2}$$

Automatic speech recognition module

ASR systems are usually trained on general-purpose speech datasets. This could not give accurate results on ATC speech due to its unique characteristics as they use strictly defined grammar, accented English, and suffer from high background noise. To solve such problem, in our framework, we utilize the XLSR–Wav2Vec2 model to convert the speech into text. XLSR–Wav2Vec2 has many advantages including over the traditional models. XLSR–Wav2Vec2 is robust as it is pre-trained on a large size dataset of diverse languages (including accented English), demonstrating robustness when fine-tuned for ATC tasks. Fine-tuning a pre-trained model like XLSR–Wav2Vec2 is not computationally expensive compared to training a deep learning model from scratch. This is essential in such applications which have limited resources. Fine-tuning XLSR–Wav2Vec2 on a smaller amount of ATC-specific data can achieve performance comparable to traditional ASR systems that require more training data.

As shown in Fig. 4, XLSR–Wav2Vec2 uses a combination of convolutional layers, vector quantization, transformers, and a masked language modeling objective to

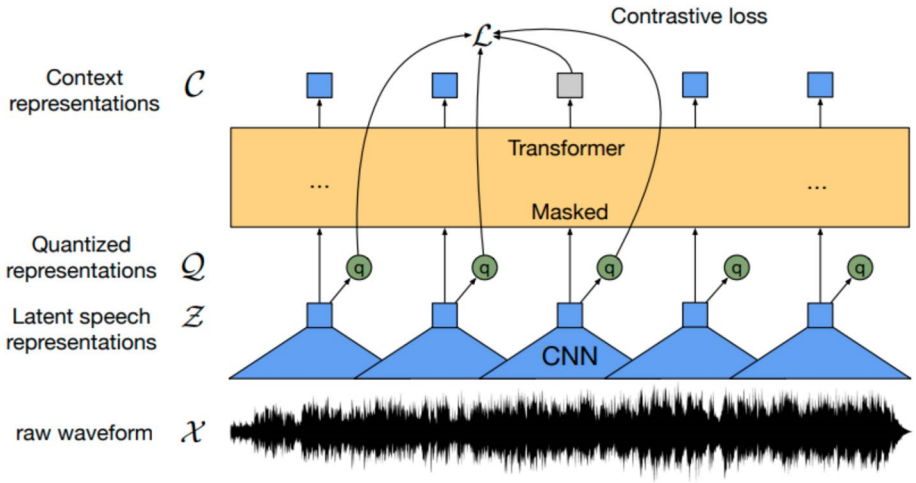


Fig. 4 XLSR–Wav2Vec2 model architecture for automatic speech recognition

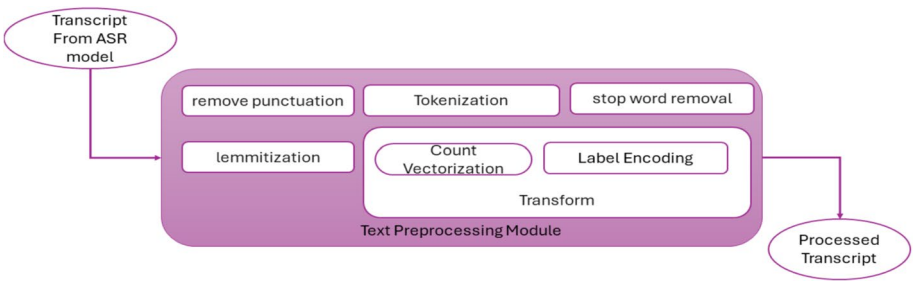


Fig. 5 Workflow for the text preprocessing module

efficiently extract informative features from raw audio data and learn a high-level representation of speech for tasks like ASR. The audio waveform, represented as a digital signal, is fed into the model.

Text preprocessing module

Text preprocessing module plays an important role in ATC transcription by providing communication data for subsequent analysis and processing. ATC conversations may contain background noise and extraneous sounds. Text preprocessing can eliminate punctuation, nonverbal sounds, and potentially even silence markers. This reduces irrelevant information and focuses the data on the core speech elements. It is responsible for preparing raw texts for the deep learning classification model by performing data cleaning and conversion processes. Data cleaning includes various tasks like removing punctuation, splitting text into tokens (words or meaningful units), and removing stop words. Additionally, it performs lemmatization or stemming to capture the core meaning of words. Finally, the preprocessed text is converted into numerical representations (vectors) using count vectorization and label encoding. This representation allows deep learning model to understand and process the text data effectively as shown in Fig. 5.

Keywords or short words significant to ATC communication are recognized by the preprocessing module (e.g., "cleared for takeoff," "wind shear alert"). These keywords might then be subjected to count vectorization to produce features that represent how frequently they appear in each chat. For some operations in the ATC analysis pipeline, label encoding may be pertinent. For instance, to classify conversations based on the meaning, the labeling then uses label encoding to convert those labels (e.g., "*Radar Contact*" → 0, "*Climb Instruction*" → 1) for the Neural network classification model.

Natural language understanding module

The natural language understanding module is responsible for understanding the text produced from the speech recognition module. The general architecture of the NLU module is shown in Fig. 6. The module contains three main stages: context-based keyword extraction, which focuses on identifying the main parts of the ATC instruction, extracting meaningful keywords for classification, and labeling which is used in extracting the important keywords from the instruction text. These keywords are considered indicators and enables correct categorization of the instructions.

Deep learning text classification model is employed for automated classification which is trained on large set of pre-labeled instructions to categorize new instructions [33]. It used in extracting the meaning from the conversations by identifying the conversation purposes and labeling, and classification of instruction based on the keywords extracted.

Keyword extraction is a method of detecting the important words in an instruction to make a classification system. This phase involves the identification of the keywords of the ATC instruction (excluding the callsign and flight number) and then categorizing the instruction. It includes the following steps:

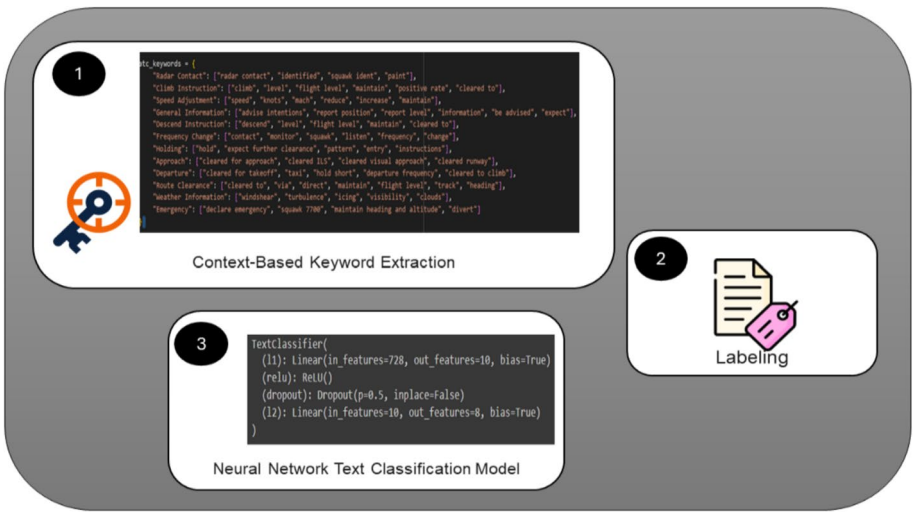


Fig. 6 Workflow for the natural language understanding module

Instruction breakdown

The instruction is broken down into a Callsign**,** which is the identifier of the aircraft, usually composed of the airline identifier (e.g., "*Sabena*") and the flight number (e.g., "*nine seven zero*"). The callsign represents the aircraft in communications, such as "*Sabena nine seven zero*." In some cases, the callsign may be the name for specific airlines company, like "*Speedbird*" representing British Airways. Follows is the instruction body which is the essential part of the message containing the action or command for the pilot. It includes the commands to be performed such as "*climb to flight level three four zero*," which means ascending the aircraft to a specific altitude.

For example, an ATC command might be: *"Sabena nine seven zero, climb to flight level three four zero."* In this case, "Sabena nine seven zero" is the callsign, whereas "climb to flight level three four zero" is the instruction body.

Leveraging prevalent keywords

Our study of the dataset proved that most of the ATC instructions are based on a set of important keywords that are used to give their meaning. In this step, we can discover the meaning of these keywords thus enabling the understanding of the instruction.

Our system focuses on the instruction body to extract only the keywords that reveal the meaning of the instruction. These keywords act like indications which helps classifying the instruction into the proper category, for example, in the sentence "*climb to flight level two nine zero*," the word "*climb*" is the keyword that shows that the instruction represents a "*Climb Instruction*" category. In this manner, the system identifies and thus comprehend that this keyword is a signal for the aircraft to increase its altitude.

ANN is employed in our system to categorize ATC instructions by learning from the large amount of labeled data. The NN model architecture is shown in Fig. 7 which is composed of an input layer (728 inputs) followed by two hidden layers. The first hidden layer has 10 neurons while the second layer contains 8 neurons where ReLU activation function is used. The output layer has 8 neurons corresponding for the 8 outputs representing the 8 text classification classes.

The input layer receives the preprocessed text data as a numerical vector of size d = 728. This vector represents the features extracted from the text, and each element x_i

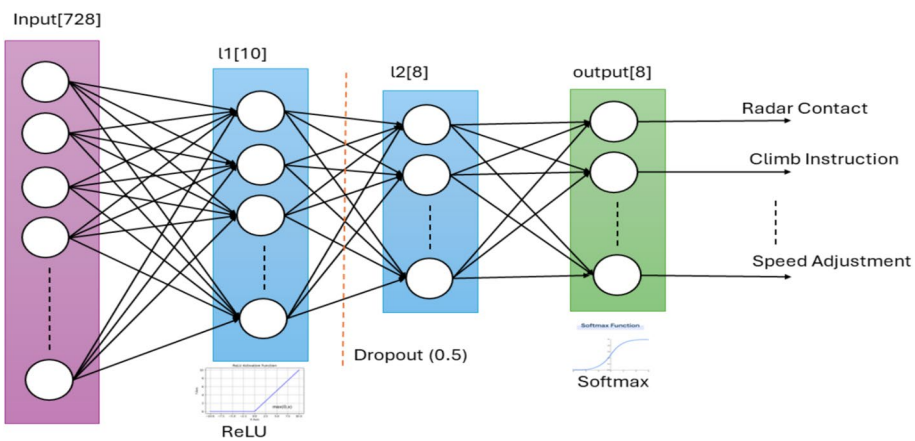


Fig. 7 Architecture of the natural language understanding model

(where i ranges from 1 to d) corresponds to a specific feature. This is followed by the first linear layer that performs a linear transformation on the input vector X = [x_1, x_2, …, x_d]. It has 10 output features, meaning there are 10 neurons in this layer.

The ReLU (rectified linear unit) activation function is applied element-wise to the outputs (z_j) from the previous layer. ReLU sets any negative value to zero and keeps positive values. A dropout technique is applied with a probability $p = 0.5$.

The second linear layer receives the output from the previous layer. Finally, the output layer contains 8 neurons (corresponding to the 8 output features from l2) and apply a SoftMax activation function to convert the outputs into probability distributions across the possible text categories. We used a cross-entropy loss which is a common loss function used in multi-class classification tasks with neural networks. It measures the difference between the predicted probability distribution (output of the network) and the true probability distribution (representing the actual class labels). The cross-entropy loss for a single data sample is calculated as in (3):

$$Loss = -\Sigma\left(t_i * log\left(y_i\right)\right) \tag{3}$$

Natural language generation module

NLG module is responsible for generating natural language responses (i.e., pilot responses) based on the information extracted by the previous module from the conversation. It receives the processed information including: extracted keywords and identified conversation subject (e.g., rise request). Based on this, it generates a natural language response appropriate to the conversation subject: such as a pilot's reply acknowledging a clearance, or a request for clarification. The overall system is shown in Fig. 8.

To prepare the text sequence for neural network processing, sequence padding is used as neural networks cannot operate on unequal sequence lengths as they result in matrices with different dimensions, which the network cannot process. The padding is used to make sure that the sequences have the same length, thus they are batched together and

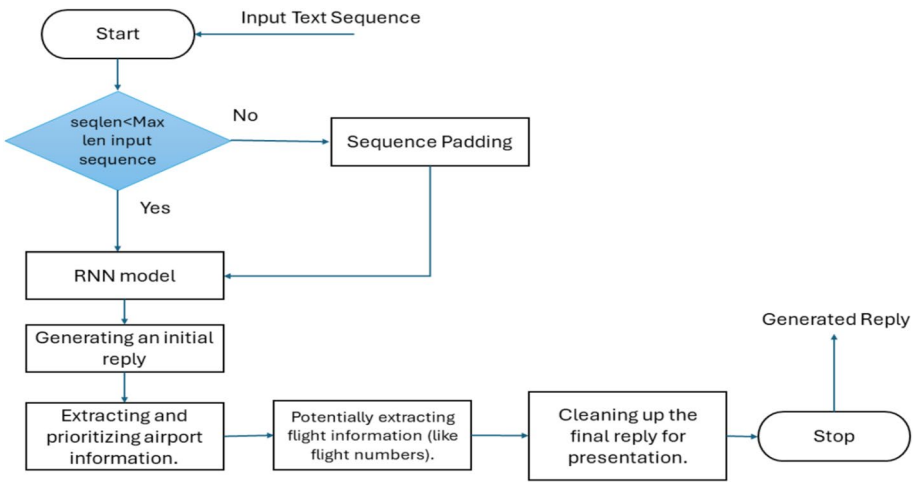


Fig. 8 Flowchart of the natural language generation model

fed into the model efficiently. The padding is implemented by adding padding characters "zero" to the end of shorter sequences to make them all have the same length.

A text generation model uses pre-trained Keras' sequential model. The model architecture is composed of several layers. The embedding layer takes an integer representing a word (likely from the tokens' vocabulary) as input and maps it to a dense vector of a specific dimensionality. This vector captures semantic relationships between words. The LSTM layer is a long short-term memory (LSTM) unit that can handle sequential data like text. It processes the sequence of embedded words from the previous layer and learns long-term dependencies between words. The dense layer: This final layer outputs a probability distribution over the vocabulary for each word in the sequence. This indicates the likelihood of each word appearing next in the sequence.

The following algorithm is used to generate the text, which is composed of the next phases. It works as follows: It generates an initial reply from the user input using a pre-trained text generation model. Then, it extracts and prioritizes airport information by processing the generated reply, the user input, and a list of clean airport names. The flight information is extracted from the processed text, the user input, and possibly a list of numbers as input and extracts flight information (potentially flight numbers) from the generated response. The reply is cleaned up using the processed text (outp3) by removing unnecessary prepositions, conjunctions, and other short words, generating the final reply.

Results and evaluation

As the proposed work constitutes many building blocks (i.e., modules), each of them has a different objective, different datasets, and different performance parameters, the reported results will be divided into subsections to test each module's performance. First, we will define the environment in which the results were obtained followed by the used datasets for the various modules, and finally the results and their analysis.

Experiments were conducted on a Lenovo IdeaPad 5 equipped with an 11th Gen Intel(R) Core(TM) i7-1165G7 @ 2, 80GHz, NVIDIA GeForce MX450 GPU, and Intel Iris Xe graphics. Google Colab is used for training and experimentation purposes via Colab runtime with GPU acceleration.

Algorithm Text Generation Model based on (RNN)

Inputs:
Set of user input, a pre-trained text generation model

Output:
Generated Text

// Generating an Initial Reply

1. Prepares the input text by converting it to a sequence of integers using the tokenizer.
2. Pads the sequence to a fixed length to ensure compatibility with the model.
3. Feeds the padded sequence into the model and obtains the model's prediction, which is a probability distribution over the vocabulary for the next words in the sequence.
4. Applies temperature scaling to adjust the randomness of the generated text.
5. Performs random sampling based on the predicted probabilities to select the next words, building a reply sentence

// Extracting and Prioritizing Airport Information:

6. Iterates through the list of airport names.
7. Use regular expression to find matches for the airport name in both input text and output text which is generated reply.
8. If a match is found in the user input, it extracts the airport name, removes it from the generated reply, and reformats the reply to place the airport information first.
9. If a match is found only in the generated reply, it treats that information as prioritized and returns the reply with the airport name at the beginning.
10. If no airport name is found in either text, it returns the original generated reply unchanged.

// Extracting Flight Information:

11. Define a regular expression pattern to match sequences of written-out numbers (e.g., "one two three").
12. Search for matches of the number pattern in the user input. This helps identify potential flight numbers mentioned by the user.
13. Iterate through the list of known flight numbers.
14. For each known number, it converts them to a string and searches for a match in the generated response.
15. If a known flight number is found in the generated response:
16. Remove unnecessary whitespace with another regular expression.
17. Clean the text by removing extra spaces and potentially punctuation
18. If flight numbers were found in the user input:
19. Splits the generated response at commas to potentially separate flight information.
20. Constructs a new string by combining the first part of the split response, the first found flight number from the user input, and the second part of the split response.
21. If the flight number is not found (either in known numbers or user input), keep the original generated response.

// Cleaning Up the Reply:

22. Define a regular expression pattern to match specific words or phrases that might appear at the end of a sentence (e.g., "*for*", "*by*", "*the*", "*thank*").
23. Iterate up to y times (which is an adjustable parameter) to attempt removing ending patterns.
24. If a match is found, remove the matched ending pattern and strips any remaining whitespace.
25. If no ending pattern is found after an iteration, the loop breaks, and the cleaned text is returned.
26. **// Generating Final Reply:**
27. Call generate_reply to get an initial response from the text generation model.
28. Feed the generated reply and the user input to airport_first to prioritize airport information.
29. Pass the processed text from the previous step (potentially containing extracted flight information) to a function to extract flight details.
30. Perform final cleaning of the reply using the strip_ending function.
31. Return output repl

Used datasets

Several datasets are used in this framework to train the deep learning models employed in the framework's modules. In the next subsections, we will explain these datasets and illuminate which module will benefit from them.

Seven datasets are used to train the automated speech recognition model (ASR) in the ATC communication module [44]. The datasets are in English with various accents, speech rates, and data quality. Two private databases: NATS (London Approach. They are manually transcribed, good quality speech (8 kHz). It contains 34 h (32 h of training, and 2 h of testing). LiveATC-Test dataset which is gathered from publicly

accessible channels (ATCO2 project). It has low quality (5–15 dB SNR) speech with accents (16 kHz). Four public databases: (I) ATCO2-Test (Evaluation set) in https://www.atco2.org/data. The full corpus is available ELDA in https://www.elra.info/catalogues/, which is a mix of noisy and accented recordings from 7 airports, open-source (1.1 h), mono-channel, 16 kHz, 16-bit PCM. (II) LDC-ATCC which contains recordings from 3 airports in noisy channels, NIST Sphere format with transcripts and timings, 8 kHz, 16-bit PCM [11]. (III) UWB-ATCC contains recordings of ATC communication, free, single channel, 8 kHz [35]. (IV) ATCOSIM contains recordings from ATC simulations using headsets, free, 10 h, non-native speakers, 16 kHz (after up-sampling) [12].

The implementation of deep neural network Text Classification module and NLG module to generate pilot reply module use ATCOSIM Air Traffic Control Simulation Speech Corpus [12], a collection of speech recordings from air traffic controllers during simulated ATC operations. ATCOSIM provides 10 h of public, wideband recordings from real-time simulations. Recordings capture realistic controller speech (language use, background noise, stress). Utterances are segmented and orthographically transcribed. Ten controllers (German/Swiss) participated, speaking in their native tongues. Recordings were made from direct microphone headsets during simulations. A single transcriber ensured consistency using a special tool (Table Trans). Transcription rules covered standard English, International Civil Aviation Organization (ICAO) phraseology, and technical notations which makes it suitable for training our models. The dataset has 10,000 samples, each with the proper instruction format as described earlier in the instruction breakdown section, described visually in Fig. 9.

Evaluation of speech processing module

The speech processing module employs the spectral subtraction method, which involves several steps in removing background noise from an audio input. The noisy voice stream is taken as input then is transformed to the frequency domain via a Fourier transform. A noise model is developed to identify and remove the noise power spectrum from the original signal, effectively eliminating noise in the frequency domain. Finally, the clear signal is returned back to the time domain in order to recover any phase information may be lost during the noise removal step.

The evaluation in this module is made in terms of MSE, SNR, and spectral convergence. In the parts of the experiment, we perform experiments with different α values

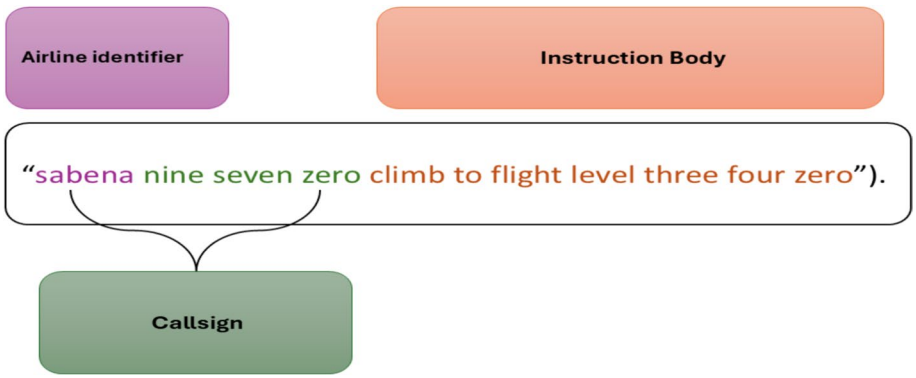


Fig. 9 Air traffic control instruction format

Table 2 Performance of the speech processing module

	Without	MSE after using Spectral Subtraction	SNR after using spectral subtraction	Spectral convergence after using spectral subtraction
Reduced background noise	Background remains same	–	Higher SNR indicates greater noise reduction	–
Enhanced communication clarity	Communication clarity is potentially hampered	–	Communication clarity is improved as background noise is reduced	–
Convergence level between estimated noise spectrum and actual noise	No spectral subtraction	–	–	Slight decrease in SC when **α** increases, and better spectral match at lower **α**
Potential for distortion	There is no distortion	Low MSE (minimum distortion)	Increased SNR implies distortion, especially at higher **α** values	–

Table 3 Results of speech processing module at different values of α

Optimization for ATC environments	MSE after using spectral subtraction	SNR after using spectral subtraction	Spectral convergence after using spectral subtraction
α = 0.1	3.1334737e-12	5.813494920730591	0.51206520
α = 0.2	2.9954070e-12	5.838876366615295	0.51057110
α = 0.3	2.9313715e-12	5.726442337036133	0.51275840
α = 0.4	2.9132122e-12	5.726442337036133	0.51722306

(0.1–0.4) to find the optimal balance between noise reduction and speech clarity for ATC communication as shown in Table 2. The value used for alpha is 0.3, which is in safe starting point to avoid introducing artifacts. By listening to the denoised signal at different alpha values, the value = 0.3 offers a suitable balance between noise reduction and signal preservation. Tables 2 and 3 summarize the effects of spectral subtraction, the noise reduction technique, on various aspects of audio quality. The leftmost column outlines what happens without spectral subtraction, while the rightmost column details the potential impacts when it is applied.

A low MSE indicates low distortion is caused by the noise reduction process. While, higher SNR indicates successful noise removal, meaning the desired speech signal has gained strength relative to the background noise. We can measure spectral convergence to determine the ability of the estimated noise spectrum to match the actual noise present in the audio. A higher convergence value implies a better match and thus more effective noise reduction. The results in Table 3 present the performance of the speech processing module under different values of α. We also use key metrics for evaluation including mean squared error (MSE), signal-to-noise ratio (SNR), and spectral convergence (SC).

Table 3 show that MSE consistently decreases as α increases from 0.1 to 0.4, with the lowest error obtained at $\alpha = 0.4$ (2.9132122e-12). This indicates that higher α values improve spectral subtraction accuracy, leading to more precise noise reduction. In terms

of SNR, it remains almost stable at all values having ranges between 5.72 and 5.83 dB. The highest SNR value = 5.8389 dB obtained at $\alpha = 0.2$. This value can be used to give balance between noise suppression and signal clarity.

Spectral convergence (SC) improves when α increases and it has maximum value (SC = 0.5172) at $\alpha = 0.4$. As SC measures indicates the speech spectrum alignment with the actual speech signal, we can conclude that higher α values lead to a more accurate restoration of speech signals.

As a conclusion, we can say that: While $\alpha = 0.4$ achieves the lowest MSE and highest SC, $\alpha = 0.2$ provides the highest SNR, which is crucial for speech intelligibility in ATC environments. Since SNR slightly decreases at $\alpha = 0.3$ and $\alpha = 0.4$, further testing with smaller step size increments (e.g., 0.25 or 0.35) could help refine the optimal α value. Depending on whether speech clarity (SNR) or noise suppression (MSE/SC) is prioritized, $\alpha = 0.2$ or $\alpha = 0.4$ would be the preferred choice for deployment in real-time ATC communication systems.

Evaluation results for automatic speech processing module

We compare our results to the recent work introduced by Zuluaga-Gomez et al. [44, 45], which used pre-trained model in ASR domain. In their work, they deduced that E2E models, Wav2Vec 2.0 and XLSR, significantly outperform the hybrid-based model on all four datasets mentioned in Sect. 5.2.1, even when fine-tuned on less data (32 h vs. 132 h). The relative word error rate reduction (WERR) achieved by E2E models ranges from 19 to 41% on challenging test sets (ATCO2-Test and LiveATC-Test). The evaluation in this module is made in terms of Micro-F1.

More specific real-life data (132 h) is added to the fine-tuning procedure to reduce WERs in clean test sets (NATS and ISAVIA). However, performance on difficult, partially out-of-domain test sets (ATCO2-Test and LiveATC-Test) increases dramatically with additional fine-tuning data. On challenging test sets containing accented English speech, E2E models pre-trained on multilingual data (w2v2-XLSR+) outperform single-language pre-trained models (w2v2-L-60 k) by a slight but noticeable margin.

E2E models achieve acceptable WERs even with limited fine-tuning data (as low as 100 utterances or 5 min of audio). Scaling up the fine-tuning data to 8 h (around 8000 utterances) leads to a significant WERR (around 50%) on private test sets (NATS and ISAVIA). E2E models fine-tuned on one ATC corpus might not transfer well to different ATC corpora. For example, the UWB-ATCC corpus transferred better to other databases compared to the LDC-ATCC corpus. Overall, the results demonstrate that E2E ASR models are a promising alternative to traditional hybrid-based models for ATC speech recognition, especially for challenging and out-of-domain scenarios.

Our results as shown in Table 4, the ATCOSIM dataset is used for test purpose, including 6400.wav files of simulated ATC communications. The results demonstrate the efficacy of our model, with an overall accuracy in terms of word level (which computes the number of correctly identified words) = 91.73, indicating a high rate of correct recognition. Additional performance metrics were also evaluated: precision = 91.98, reflecting the proportion of true positive results among all positive results predicted by the model; recall 91.59, representing the ability of the model to correctly identify all relevant instances; and F1 score = 91.74, providing a balanced measure of precision and recall.

Table 4 Performance results of the recognition module ATCOSIM at different numbers of files

Test Data (~ 6 h)	Accuracy	Precision	Recall	F1 score	WER	SER
100 files	0.9135	0.9180	0.9135	0.9147	0.1367	0.3352
200 files	0.9229	0.9301	0.9209	0.9217	0.1189	0.3651
500 files	0.9134	0.9183	0.9148	0.9182	0.1381	0.3860
1000 files	0.8962	0.8982	0.8962	0.8923	0.1701	0.4093
1500 files	0.8889	0.8902	0.8962	0.8983	0.1803	0.4203
2000 files	0.8934	0.8962	0.8942	0.8913	0.1731	0.4003
2500 files	0.8912	0.8926	0.8952	0.8953	0.1791	0.4034
3000 files	0.9055	0.9048	0.9025	0.9047	0.1467	0.3852
3500 files	0.9017	0.9049	0.9016	0.9023	0.1582	0.3822
4000 files	0.9038	0.9072	0.9038	0.9045	0.1542	0.3825
4500 files	0.9056	0.9090	0.9056	0.9063	0.1502	0.3694
5000 files	0.9075	0.9119	0.9075	0.9082	0.1465	0.3564
5500 files	0.9095	0.9133	0.9098	0.9104	0.1428	0.3481
6000 files	0.9119	0.9164	0.9125	0.9131	0.1390	0.3401
6200 files	0.9129	0.9170	0.9126	0.9138	0.1381	0.3382
6400 files	0.9173	0.9198	0.9159	0.9174	0.1361	0.3327

Table 5 Accuracy of NLU Module in different preprocessing tasks

Preprocessing tasks	Max. Micro-F1
Before preprocessing	0.9762
After preprocessing 3 steps (tokenization–punctuation removal–stop word removal)	0.9772
After preprocessing 4 steps (tokenization–punctuation removal–stop word removal–stemming)	0.9802
After preprocessing 5 steps (tokenization–punctuation removal–stop word removal–lemmatization) (proposed one)	0.9816

The word error rate (WER) = 13.61, showing the percentage of words incorrectly predicted, and the sentence error rate (SER) = 33.27, indicating the proportion of sentences with at least one error. These metrics highlight the robustness and reliability of our ASR system, showcasing its potential to enhance the operational efficiency and safety of ATC systems.

Evaluation of the text preprocessing module

Text preprocessing improves the quality of ATC transcription data by focusing on word meaning and creating a format suitable for further analysis in the NLU model. The steps used to improve the quality are: punctuation removal, tokenization, stop word removal, lemmatization, and transformation.

Table 5 shows the impact of the different text preprocessing tasks on the F1 score parameter. The table presents the maximum Micro-F1 score achieved for each preprocessing configuration. Micro-F1 is a common metric used to evaluate the overall performance of a classification model, considering both precision and recall. The first row shows the baseline performance, where the NLU module is applied directly to the text data (before preprocessing) and achieves a Micro-F1 score of 0.9762. The effect of each preprocessing step can be summarized as follows. Tokenization, punctuation removal,

and stop word removal achieve a higher Micro-F1 score of 0.9772, adding minor improvement. The stemming leads to a more significant increase in score, reaching 0.9801. The lemmatization (proposed, the final row in the table, which includes tokenization, punctuation removal, stop word removal, and lemmatization) adds to the overall results improvement. The highest Micro-F1 score of 0.9816, indicating the most significant improvement in NLU module accuracy.

Evaluation of natural language understanding module

The context-based keyword extraction approach has exhibited good results in finding important parts for ATC instruction classification. The system divides the ATC instructions into three main components: callsign, flight number, and instruction body. This separation helps to focus on the most important instructional content for classification. These keywords serve as markers of instruction type, allowing the system to efficiently categorize them. These keywords are fed the ANN model allowing the ANN to learn more effectively and achieve better classification accuracy.

It is important to obtain robust context-based keyword extraction which will impact the performance of the overall NLU module. This method improves the classification accuracy by providing an improved and instructive feature set for the ANN model. The F1 score reaches 98%. It also reduces noise and ambiguity in the data by separating the instructions from the non-important parts such as callsigns, allowing the ANN to focus on the main instructional information.

Evaluation of natural language generation module

The NLG module effectively uses pre-trained text generation models to create responses tailored to the application at hand context. This module is evaluated using several metrics. BLEU is used to evaluate the quality of generated texts by comparing them against the original texts (takes values from zero to one). ROUGE is used to evaluate the similarity between the generated texts and the original texts. METEOR: used to evaluate the quality of the generated texts based on word alignment and meanings.

As shown in Table 6, compared to traditional ATC systems, the proposed NLG module demonstrates significant improvements in response times and pilot workload reduction. Initially, over 10 test requests. The module achieved an average latency of 0.60 s, swiftly responding with an impressive throughput of 1.66 requests per second. Notably, both the 90th percentile latency of 0.63 s and the 99th percentile latency of 0.63 s indicate consistent performance under varied loads, with zero errors detected, ensuring reliability and robustness. This capability enhances communication efficiency, effectively minimizing pilot workload and optimizing air traffic management. The obtained

Table 6 Response time and throughput for generating pilot replies at different numbers of requests (10 and 100)

Total requests	Successful requests	Error count&error rate	Average latency	90th Percentile latency	99th Percentile latency	Throughput
10	10	0,0.0	0.60	0.63	0.63	1.66
100	100	0,0.0	0.61	0.649	1.008	1.629

Table 7 Word error rates (WERs) in percentages (%) evaluated in ATCOSIM Test set on different systems

System	Train set	Test set LM-3 LM-4	
TDNF-B Zuluaga-Gomez et al. (2023) [44]	OOD set	67.5	63.4
TDNNF-TF-B Zuluaga-Gomez et al. (2023) [44]	Train1	7.5	7.3
TDNNF-TF-B Zuluaga-Gomez et al. (2023) [45]	Train2	23.9	20.5
TDNNF-TF-B Pan et al. (2023) [29]	TR1 + Tr2	5.9	6.0
TDNNF Pan et al. (2023) [29]	Train1	8.5	8.1
CNN + TDNNF Ardila et al. (2020) [3]	TR1 + Tr2	5.0	5.1
TDNNF-BPE Ardila et al. (2020) [3]	TR1 + Tr2	LM-6—> 7.2	

Table 8 Error rates on the ATCOSIM test set

System	Train set	Gender	ATCOSIM Test set
w2v2-L-60 k Zuluaga-Gomez et al. (2023) [44]	LDC-ATCC	(Male)	41.1
w2v2-L-60 k Zuluaga-Gomez et al. (2023) [45]	LDC-ATCC	(Female)	35.7
Proposed ASR	UWB-ATCC	(~6 h -mixed)	13.61

module results after performing the experiments are BLEU = 0.85, ROUGE = 0.80, and METEOR = 0.78. Expanding the evaluation to 100 requests, the NLG module continues to excel, achieving an average latency of 0.61 s and a throughput of 1.629 requests per second. The 90th percentile latency of 0.649 s and the 99th percentile latency of 1.008 s maintain the system's consistency in handling peak demands, underscoring its reliability with an error rate of almost 0. These metrics underscore the NLG module's capacity to sustain high-performance standards, offering dependable and efficient communication crucial for enhancing safety and operational efficiency in air traffic control systems.

Overall system results and analysis

Tables 7 and 8 demonstrate the results for various systems such as TDNF-B, TDNF-TF-B, TDNNF, CNN + TDNNF, and TDNNF-BPE, showing the error rates on the ATCOSIM test set. For example, CNN + TDNNF trained on TR1 + Tr2 achieves the lowest error rates of 5.0 and 5.1 on LM-3 and LM-4, respectively. Table 8 focuses on the w2v2-L-60 k system evaluated on the LDC-ATCC and UWB-ATCC datasets by gender, revealing that female speakers generally produce lower error rates compared to male speakers. The proposed ASR system, tested on the mixed-gender UWB-ATCC dataset, achieves a much lower error rate of 13.61. Together, these tables highlight the variation in speech recognition performance across different paradigms, training groups, and speaker characteristics. It depicts word error rates (WERs) in percentages (%) evaluated in the ATCOSIM test set on different systems.

Comparison with traditional ATC systems

Table 9 provides a comparative analysis of the traditional air traffic control (ATC) systems versus the proposed AI and NLP-enabled ATC systems. This table highlights the key areas where AI and NLP can significantly improve upon traditional

Table 9 Comparative analysis of traditional vs. proposed AI and NLP-Enabled air traffic control systems

Aspect	Traditional ATC systems	Proposed AI and NLP-enabled ATC system
Data processing	Reliant on human operators, limited processing capabilities	Analyzes real-time flight data. Optimizes airspace utilization and minimizes congestion
Decision support	Relies on human judgment, prone to errors	Provides valuable insights and recommendations for informed decision-making
Communication	Standardized protocols, potential for misinterpretation	Interprets natural language instructions, enhances clarity, and reduces misinterpretation
Predictive maintenance	Reactive maintenance approach	Proactive solutions anticipate and address potential issues before escalation
Efficiency and accuracy	Relies on human operators prone to errors due to fatigue, accents	Automates transcription, reducing errors and enhancing accuracy Improves communication accuracy by up to 25%
Response time	Manual processing leads to slower response times during peak traffic. Extensive and resource-intensive training required for operators	Real-time processing allows for quicker responses to situational Changes. Requires less human input and adaptability through machine learning
Scalability and adaptability	Require physical upgrades and modifications	Can be continuously updated and improved through software updates
Benefits	Improved efficiency, safety	Enhanced operational efficiency, communication clarity, safety, and reliability

methods, NLP-enabled ATC. Overall, this research demonstrates the potential of AI and NLP to significantly improve efficiency, safety, and communication within air traffic management.

The proposed ASR system demonstrates strong performance with an error rate of 13.61 when tested on the ATCOSIM dataset (approximately 6 h of mixed data). While other models, particularly those utilizing combined datasets and complex architectures like CNN + TDNNF, achieved lower error rates, the proposed ASR system's efficiency with the provided test data is notable. This comparison underscores the effectiveness of the proposed ASR system in scenarios where the test data is limited, showcasing its potential as a robust and versatile speech recognition solution.

Finally, Table 10 depicts the quantitative comparison with the most recent existing research versus the proposed solution [3–5, 15, 20, 22, 29, 37]. The table highlights that while standard BERT provides a strong baseline, combining BERT with additional layers like TextCNN, GRU, and BiGRU can further enhance accuracy. The EBA and the proposed NLU models achieved the highest accuracy (0.982).

Discussion

This study has comprehensively evaluated the performance of a proposed AI-enabled ATC communication system composed of several modules: automatic speech recognition (ASR), natural language understanding (NLU), and natural language generation (NLG). Several experiments were conducted evaluating each individual component using multiple performance metrics across modules to ensure robustness and real-time efficiency, then the overall system. Through these experiments, we have chosen the best values for noise reduction to attain good results by the automatic ASR

Table 10 Evaluation metric for different intent-based natural language understanding model

Model	Accuracy
BERT Ardila et al. (2020) [3]	0.957
BERT-Jieba tokenizer Pan et al. (2023) [29]	0.911
BERT-LSTM Lin et al. (2020) [20]	0.956
BERT-BiLSTM Lu et al. (2020) [22]	0.951
BERT-TextCNN ArunKumar et al. (2022) [4]	0.96
BERT-GRU Kim (2014) [15]	0.968
BERT-TextCNN-BiGRU Chandra et al. (2023) [5]	0.975
EBA Tikayat Ray et al. (2023) [37]	0.982
Proposed NLU	~0.982

module. The spectral subtraction method was assessed using MSE, SNR, and spectral convergence (SC). The results showed that $\alpha = 0.3$ was selected as a practical balance between SNR and SC. The ASR model was trained and tested on the ATCOSIM dataset, where it achieved 91.73% word-level accuracy, with a word error rate (WER) of 13.61% and an F1 score of 91.74, showing its reliability for ATC communication. Compared to state-of-the-art models like Wav2Vec 2.0 and XLSR, our system performs competitively—especially under limited training data—demonstrating scalability and adaptability.

The text preprocessing (including lemmatization and stop word removal) improved Micro-F1 scores significantly, with the best score of **0.9816 in the NLU module**. Context-based keyword extraction further enhanced the classification accuracy (F1 = **98%**) by concentrating on the key instruction elements and reducing the irrelevant information, thus improving the model generalizability and precision. As for the NLG module, we were able to maintain high performance with BLEU = 0.85, ROUGE = 0.80, and METEOR = 0.78. We also demonstrated that we can give fast response times (~ 0.60 s), low latency even under load, and zero error rates, underscoring its reliability in real-time ATC settings.

As an overall system evaluation, the comparative analysis against several existing models (e.g., CNN + TDNNF, BERT + TextCNN) shows that some models achieved lower WERs than our system on larger or combined datasets, however, our proposed system remains highly effective on constrained ATC datasets. The proposed AI and NLP-powered ATC communication system exhibits strong performance across all modules—offering accurate, efficient, and real-time support for air traffic operations. Each module is optimized for specific tasks, contributing to a cohesive pipeline that significantly reduces human error, enhances speech recognition accuracy, improves communication clarity, and ensures scalability. The following results are important from the experiments worth mentioning. In **Noise Reduction,** best values of $\alpha = 0.2–0.4$, while $\alpha = 0.3$ provides a balance between clarity and suppression. **The ASR Module** exhibited 91.73% accuracy, 13.61% WER; performs well on simulated ATC data. **In the NLU Module, the** preprocessing and keyword extraction raise F1 score to **0.9816.** The **NLG Module** is fast and robust with BLEU = 0.85 and latency ~ 0.6 s. **The overall system** competes effectively with recent models, offering practical performance even with limited training data.

Conclusion

The role of natural language processing (NLP) in air traffic control (ATC) systems marks a significant paradigm shift in aviation operations. As demonstrated through various case studies and comparative analyses, these advanced technologies offer substantial improvements over traditional systems across multiple facets of ATC operations, including data processing, decision support, communication, maintenance, efficiency, accuracy, response times, scalability, and overall operational benefits.

AI and NLP technologies not only enhance the capacity and efficiency of ATC systems but also significantly increase the safety and reliability of air traffic management. By automating complex processes and reducing the reliance on human operators, who are prone to fatigue and error, these technologies minimize the risk of miscommunication and operational mistakes, which are critical in high-stakes environments like aviation.

Furthermore, the adaptability and scalability provided by AI and NLP systems mean that ATC can continue to evolve with the increasing demands of global air traffic without the constant need for extensive physical upgrades. Software updates and machine learning algorithms allow these systems to adapt to new challenges and learn from real-time data, ensuring that improvements are continuous and aligned with current needs.

However, the transition to AI and NLP-enhanced systems is not without challenges. It requires significant investment in technology, training, and system redesign, as well as ongoing maintenance and updates to ensure the systems remain effective and secure against emerging threats. The cultural shift for ATC operators and stakeholders toward trusting automated systems also plays a crucial role in the successful implementation of these technologies.

In conclusion, while the path to fully integrated AI and NLP ATC systems involves careful planning, significant resources, and continuous improvement, the potential benefits justify the efforts. Enhanced efficiency, improved safety, reduced operational costs, and better scalability promise to revolutionize air traffic management. As these technologies mature and their applications within ATC systems become more refined, the future of aviation looks increasingly safe, efficient, and adaptable to the growing demands of global air traffic.

Author contributions

Amany M. Sarhan and Hesham A. Ali conceptualized the study, added the basic ideas, wrote and edited the manuscript, and were involved in supervision. Amany M. Sarhan and Rawda Fathy were responsible for formal analysis, methodology, software, validation, and writing—original draft preparation. Amany M. Sarhan, Rawda Fathy, and Hesham A. Ali assisted with investigation, methodology, and validation.

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Data availability

The dataset used in this paper is publicly available online.

Declarations

Conflict of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

1. Alla H, Balouki Y, Moumoun, L (2023) The use of chatbots as supportive agents in air transportation systems. Proceedings of the international conference on intelligent system and smart technologies, Cham: Springer International Publishing, 239–250, Jan. 2023
2. Aman E, Jana S, Athikary KG, Suryanarayana RC (2023) AI Inspired ATC, based on ANN and Using NLP, SAE Technical paper, No. 2023–01–0985, 2023
3. †Ardila R, Branson M, Davis K, Henretty M, Kohler M, Meyer J, Weber G (2020) Common voice: a massively-multilingual speech corpus. Proceedings of the 12th conference on language resources and evaluation (LREC 2020), 4218–4222, May 2020
4. ArunKumar K, Kalaga DV, Kumar CMS, Kawaji M, Brenza TM (2022) Comparative analysis of Gated Recurrent Units (GRU), Long Short-Term memory (LSTM) Cells, Autoregressive Integrated Moving Average (ARIMA), Seasonal Autoregressive Integrated Moving Average (SARIMA) for forecasting COVID-19 trends. Alex Eng J 61:7585–7603
5. Chandra C, Jing X, Bendarkar MV, Sawant K, Elias L, Kirby M, Mavris DN (2023) Aviation-BERT: A preliminary aviation-specific natural language model. Proceedings of the AIAA AVIATION 2023 forum, San Diego, CA, USA, 12–16 June 2023
6. de Almeida EE, Oliveira AVM (2023) An econometric analysis for the determinants of flight speed in the air transport of passengers. Sci Rep 13:4573
7. Dua M, Bhagat B, Dua S, Chakravarty N (2024) A review on gujarati language-based automatic speech recognition (ASR) systems. Int J Speech Technol 27(1):133–156
8. Fan J, Pan W (2024) Customization of the ASR system for ATC speech with improved fusion. Aerospace, MDPI 11(3):219
9. Fan P, Guo D, Zhang J, Yang B, Lin Y (2024) Enhancing multilingual speech recognition in air traffic control by sentence-level language identification. Applied Acoustics, MDPI 224:110123
10. Fragua Á, Jiménez-Martín A, Mateo A (2023) Complex network analysis techniques for the early detection of the outbreak of pandemics transmitted through air traffic. Sci Rep 13:18174
11. Godfrey JJ (1994) Air traffic control complete. Linguistic Data Consortium, Philadelphia
12. Hofbauer K, Petrik S, Hering, H (2008) The ATCOSIM corpus of non-prompted clean air traffic control speech. Proceedings of the LREC, May 2008. (available at https://aclanthology.org/L08-1507/)
13. Kalyanam K. AI/ML & NLP for UAS/Air Traffic Management (U/A-TM). In ATCA Global, Nov. 2023
14. Kistan T, Gardi A, Sabatini R (2018) Machine learning and cognitive ergonomics in air traffic management: recent developments and considerations for certification. Aerospace, MDPI 5(4):103
15. †Kim Y (2014) Convolutional neural networks for sentence classification. Proceedings of the Conference on Empirical Methods in Natural Language Processing, 2014
16. Knoblock E, Gasper M, Apaza R, Li H, Wang Z, Han R, Schimpf N, Rose N, Adams N (2022) Investigation of intelligent resource management for aviation communications. Proceedings of the 39th International Communications Satellite Systems Conference (Icssc) and colloquium, Nov. 2022
17. Kristanto A, Indriyanto T (2023) Development of communication system for UAV ground control station with ATC based on controller pilot data link communication. War Penelit Perhub 35(2):243–256
18. Krivonos PD (2007) Communication in aviation safety: lessons learned and lessons required. Proceedings of the regional seminar of the australia and new zealand societies of air safety investigators 4:1–35
19. Lázaro FL, Madeira T, Melicio R, Valério D, Santos LFFM (2025) Identifying human factors in aviation accidents with natural language processing and machine learning models. Aerospace 12:106
20. Lin Y, Guo D, Zhang J, Chen Z, Yang B (2020) A unified framework for multilingual speech recognition in air traffic control systems. IEEE Transactions on Neural Networks and Learning Systems 32(8):3608–3620
21. Lin Y, Yang B, Li L, Guo D, Zhang J, Chen H, Zhang Y (2021) ATCSpeechNet: a multilingual end-to-end speech recognition framework for air traffic control systems. Appl Soft Comput 112(107847):2021
22. Lu Q, Zhu Z, Xu F, Zhang D, Wu W, Guo Q (2020) Bi-GRU sentiment classification for chinese based on grammar rules and BERT. Int J Comput Intell Syst 13:538–548
23. Luo Q, Liu H, Liu C, Deng Q (2024) Multi-strategy Cooperative Scheduling for Airport Specialized Vehicles Based on Digital Twins. Sci Rep 14:15533
24. Malakis S, Baumgartner M, Berzina N, Laursen T, Smoker A, Poti A, Kontogiannis T (2023) A framework for supporting adaptive human-AI teaming in air traffic control. Proceedings of the International Conference on Human-Computer Interaction, Cham: Springer Nature Switzerland, pp. 320–330, July 2023
25. Maynard P, Clarke SS, Almache J, Kumar S, Rajkumar S, Kemp A, Pai R (2021) Natural language processing (NLP) techniques for air traffic management planning. In AIAA AVIATION 2021 FORUM, p. 2322
26. New MD, Wallace RJ (2025) Classifying Aviation Safety Reports: Using Supervised Natural Language Processing (NLP) in an Applied Context. Safety, MDPI 11(1):7
27. Novitasari, S., Fukuda, T. and Kurata, G (2022) Improving ASR robustness in noisy condition through VAD integration. Proceedings of the INTERSPEECH, pp. 3784–3788, Sep. 2022
28. Ortner P, Steinhöfler R, Leitgeb E, Flühr H (2022) Augmented air traffic control system—artificial intelligence as digital assistance system to predict air traffic conflicts. Ai 3(3): 623–644
29. Pan W, Jiang P, Wang Z, Li Y, Liao Z (2023) Ernie-gram BiGRU attention: an improved multi-intention recognition model for air traffic control. Aerospace, MDPI 10(4):349
30. Paul S, Purkaystha BS, Das P (2018) NLP tools used in civil aviation: a survey. Int J Adv Res Comput Sci 9(2)

31. Pinska-Chauvin E, Helmke H, Dokic J, Hartikainen P, Ohneiser O, Lasheras RG (2023) Ensuring safety for artificial-intelligence-based automatic speech recognition in air traffic control environment. Aerospace 10(11):941
32. Saraf AP, Chan K, Popish M, Browder J, Schade J (2020) Explainable artificial intelligence for aviation safety applications. Proceedings of the AIAA Aviation 2020 Forum, p. 2881, 2020
33. Sarhan AM, Ayman H, Wagdi M, Ali B, Adel A, Osama R (2024) Integrating machine learning and sentiment analysis in movie recommendation systems. J Electr Syst Inf Technol 11:53
34. Sayed SA, Abdel-Hamid Y, Hefny HA (2023) Artificial intelligence-based traffic flow prediction: a comprehensive review. J Electr Syst Inf Technol 10:13
35. Šmídl L, Švec J, Tihelka D, Matoušek J, Romportl J, Ircing P (2019) Air traffic control communication (ATCC) speech corpora and their use for ASR and TTS development. Lang Resour Eval 53:449–464
36. Srinivasarao V (2023) Speech signal analysis and enhancement using combined wavelet fourier transform with stacked deep learning architecture. Int J Speech Technol 26(3):735–742
37. Tikayat Ray A, Cole BF, Pinon Fischer OJ, White RT, Mavris DN (2023) AeroBERT-Classifier: classification of aerospace requirements using BERT. Aerospace, MDPI 10:279
38. Verberk JD, van der Werff SD, Weegar R, Henriksson A, Richir MC, Buchli C, Nauclér P (2023) The augmented value of using clinical notes in semi-automated surveillance of deep surgical site infections after colorectal surgery. Antimicrob Resist Infect Control 12(1):117
39. Wang X, Mao Y, Wu X, Xu Q, Jiang W, Yin S (2023) An ATC instruction processing-based trajectory prediction algorithm designing. Neural Comput Appl 35(32):23477–23490
40. Wu Y, He S, Shi J (2024) A dynamic decision-making approach for cabin unlawful interference emergency disposal using dynamic bayesian network. Sci Rep 14:19002
41. Yang C, Huang C (2023) Natural Language Processing (NLP) in aviation safety: systematic review of research and outlook into the future. Aerospace, MDPI 10(7):600
42. Yang W, Chi Y, Huang Y, Wei W, Xu Z (2024) Layout analysis of the RCEP international airline network based on hub identification using improved contribution matrix. Sci Rep 14:10839
43. Yi L, Min R, Kunjie C, Dan L, Ziqiang Z, Fan L, Bo Y (2023) Identifying and managing risks of ai-driven operations: a case study of automatic speech recognition for improving air traffic safety. Chin J Aeronaut 36(4):366–386
44. Zuluaga-Gomez J, Prasad A, Nigmatulina I, Motlicek P, Kleinert M (2023) A virtual simulation-pilot agent for training of air traffic controllers. Aerospace, MDPI 10(5):490
45. Zuluaga-Gomez J, Nigmatulina I, Prasad A, Motlicek P, Khalil D, Madikeri S, Choukri K (2023) Lessons learned in transcribing 5000 H of air traffic control communications for robust automatic speech understanding. Aerospace, MDPI 10(10):898

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